



An Internship Project Presentation

ACTIVE LEARNING FOR PNEUMONIA DETECTION WITH VISION TRANSFORMERS AND BAYESIAN UNCERTAINTY ESTIMATION

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Agenda

- Abstract
- Terminologies Introduced
- Objectives
- Introduction
- Implementation
- Results and Discussions
- Conclusion
- References





• **ABSTRACT**

Pneumonia remains a major public health concern, particularly in vulnerable populations such as children, the elderly, and immunocompromised individuals. Early and accurate detection using chest X-rays is crucial but often limited by the availability of expert radiologists. Although deep learning models have shown promise in automating diagnosis, they typically require large labeled datasets and lack the ability to express uncertainty two critical drawbacks in clinical applications. This study presents an uncertainty-aware pneumonia detection framework that combines Vision Transformers (ViT) with Monte Carlo (MC) Dropout. ViTs effectively capture spatial dependencies in medical images, while MC Dropout enables the model to estimate predictive uncertainty by performing multiple stochastic forward passes during inference. This allows the system to identify ambiguous cases that may require expert review. To enhance data efficiency, the model also integrates an active learning approach that selects uncertain samples for annotation, thereby improving performance with fewer labeled images. Experiments on a public chest X-ray dataset show that the ViT + MC Dropout model achieves 94.1% accuracy, 96.0% sensitivity, and an AUC of 0.98, with improved uncertainty calibration 85% of predictions had over 90% confidence. Additionally, active learning further improves model accuracy using fewer annotations.

- **TERMINOLOGIES INTRODUCED**



1

VISION TRANSFORMERS

A deep learning model that applies transformer architecture to image classification tasks.

2

MONTE CARLO (MC) DROPOUT

A technique for estimating uncertainty in neural networks by performing dropout at inference time.

3

ACTIVE LEARNING

A machine learning approach where the model selectively queries the most informative data points to label.



- **OBJECTIVES**

- To develop a reliable and efficient pneumonia detection system using chest X-ray images.
- To leverage Vision Transformers (ViT) for improved spatial feature extraction over traditional CNNs.
- To incorporate Monte Carlo Dropout (MC Dropout) to estimate predictive uncertainty, enabling safer clinical decisions.
- To implement an Active Learning strategy that reduces labeling costs by selecting the most uncertain and informative samples.



• INTRODUCTION

- Pneumonia remains a serious health issue, especially for vulnerable populations.
- Chest X-rays are commonly used for diagnosis but require expert interpretation.
- Deep learning helps automate diagnosis but often lacks uncertainty estimation and needs large labeled datasets.
- The proposed framework combines Vision Transformers (ViT), Monte Carlo Dropout, and Active Learning.
- This approach improves diagnostic accuracy, uncertainty estimation, and reduces dependency on extensive annotations.



• IMPLEMENTATION

Dataset and Preprocessing:

- Used Chest X-ray Pneumonia Dataset with 5,856 grayscale images labeled as Pneumonia or Normal.
- Data split into training, validation, and test sets with class imbalance.
- Images divided into 16×16 patches for model input.

Vision Transformer (ViT) Model:

- Fine-tuned ViT Base Patch16 224 from TIMM library with modified classification head.
- Added dropout ($p=0.5$) and a two-unit linear layer for binary classification.
- Images processed as patch embeddings with positional encodings using multi-head self-attention.



- **IMPLEMENTATION**

Uncertainty Estimation & Training: Monte Carlo Dropout

- MC Dropout is applied during training and inference with dropout enabled at test time for multiple stochastic forward passes.
- Predictive uncertainty is estimated by averaging outputs and computing variance, improving reliability in ambiguous cases.
- The model uses cross-entropy loss with early stopping and data augmentation to enhance generalization.

• RESULTS AND DISCUSSIONS



- The Vision Transformer (ViT) model showed high accuracy in pneumonia detection from chest X-ray images, proving the strength of transformer architectures in medical image classification.
- By using global self-attention, the model effectively captured long-range dependencies and subtle patterns that are often missed by traditional CNNs. This ability was crucial for identifying spatially dispersed pneumonia indicators.
- Additionally, incorporating Monte Carlo Dropout during inference allowed the model to estimate predictive uncertainty, enhancing its reliability by quantifying confidence in its decisions.



***FIG 1: CHEST X-RAY IMAGE
CLASSIFIED AS PNEUMONIA***



• RESULTS AND DISCUSSIONS

METRIC	VALUE
PREDICTION	PNEUMONIA
CONFIDENCE	80.76%
CLASS PROBABILITY (NORMAL)	19.24%
CLASS PROBABILITY (PNEUMONIA)	80.76%
UNCERTAINTY	0.2084

TABLE I: VIT+MC DROPOUT RESULTS

- The model predicted the presence of pneumonia with a confidence level of 80.76%, indicating a strong likelihood of pneumonia in the input.
- It assigned 80.76% probability to the pneumonia class and 19.24% to the normal class, showing a clear inclination toward pneumonia diagnosis.
- An uncertainty score of 0.2084 further quantifies the model's doubt, offering insights into the reliability of the prediction.



• RESULTS AND DISCUSSIONS

- The comparative evaluation shows that the Vision Transformer (ViT) with Monte Carlo Dropout outperformed ResNet-50 in chest X-ray classification.
- ViT achieved 94.1% accuracy versus 92.3% for ResNet, and a higher sensitivity of 96.0% compared to 94.2%, indicating better pneumonia detection.
- The AUC was also higher for ViT (0.98 vs. 0.96). In terms of calibration, ViT had 85% correct predictions at over 90% confidence, while ResNet had only 78%, highlighting ViT's superior reliability.

METRIC	ResNet-50	ViT + MC Dropout
ACCURACY (%)	92.3	94.1
SENSITIVITY (%)	94.2	96.0
AUC (ROC)	0.96	0.98
CALIBRATION (% CORRECT >90% CONFIDENCE)	78%	85%

TABLE II: COMPARISON OF RESNET-50 AND VIT + MC DROPOUT

• RESULTS AND DISCUSSIONS



MODEL	ACCURACY
CNN BASELINE ACCURACY	72.92%
ViT BASELINE ACCURACY	87.98%
ViT ACTIVE LEARNING ACCURACY	89.26%

TABLE III: ACTIVE LEARNING RESULTS

- The results in Table III demonstrate the effectiveness of Vision Transformers (ViT) and the added benefit of incorporating active learning strategies.
- The CNN baseline model achieved an accuracy of 72.92%, while ViT achieved 87.98% without active learning. With active learning, ViT's performance improved further to 89.26%.
- This confirms that active learning can enhance model performance while reducing labeling efforts, making it highly suitable for medical imaging tasks.



• RESULTS AND DISCUSSIONS

- The confusion matrix presented in The above evaluates the classification performance of the model across three distinct classes: Bacterial Pneumonia, Normal, and Viral Pneumonia.
- The model demonstrated high accuracy in identifying Normal cases, with only 2 misclassified out of 270. However, the primary challenge lay in distinguishing between Bacterial and Viral pneumonia cases.
- The model correctly classified 436 bacterial and 183 viral cases, while 59 viral cases were misclassified as bacterial, and 82 bacterial cases were misclassified as viral.
- This overlap highlights the difficulty in distinguishing between visually similar lung pathologies.

Confusion matrix

Actual \ Predicted			
	BACTERIAL	NORMAL	VIRAL
BACTERIAL	436	3	59
NORMAL	0	268	2
VIRAL	82	10	183

FIG 2: CONFUSION MATRIX OF THE PROPOSED MODEL ON CHEST X-RAY DATASET



• **CONCLUSION**

- The proposed framework combines Vision Transformers (ViT) with Monte Carlo Dropout (MC Dropout) for effective pneumonia detection from chest X-rays.
- ViTs capture long-range dependencies and subtle patterns, while MC Dropout provides predictive uncertainty estimates, improving clinical reliability.
- Active learning further reduces the need for large annotated datasets by focusing on uncertain cases.
- The method outperforms traditional CNNs in accuracy, sensitivity, AUC, and uncertainty calibration, making it a data-efficient and interpretable solution with potential for broader medical imaging applications.



• References

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